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Top Skills

Retrieval-Augmented Generation (RAG)

C2J Modernization

Spring Boot

Certifications

Learning Impact: The Importance of Storytelling

Social Marketing Certification

Hootsuite Platform Certification

Become a CompTIA Security+
Certified Security Professional

Prabesh Thapa

Software Engineer | Java · Spring Boot · GenAI/RAG | Legacy Modernization (COBOL → Cloud) | AppSec (ex-Citi) | AWS | FinTech
Atlanta, Georgia, United States

Summary

Software Engineer with 4+ years building and modernizing enterprise systems in financial services — and using GenAI to ship that work faster without sacrificing correctness. Today at TCS (client: TSYS / Global Payments), I convert legacy COBOL into Java Spring Boot services, validate business rules with cross-team testing, and build AI-assisted workflows that cut manual code analysis from 8 hours to about 2 hours per module. I also built a RAG chat system with LangChain, embeddings, and PostgreSQL pgvector for grounded Q&A over documentation and code context. Before that, I spent 2.5 years as an Application Security Engineer supporting Citi Group: vulnerability management across 15+ application teams (+20% remediation efficiency, ~1 month faster mean-time-to-patch), Python automation that correlated NVD data with BlackDuck/Snyk findings, and Tableau dashboards that turned risk data into KPIs for 50+ applications. Earlier, I shipped full-stack apps (React, Node.js, Express, MongoDB), CI/CD with Jenkins/Docker/Maven to AWS EC2, and automated testing with Selenium and Postman. Core skills: Java, Spring Boot, Python, JavaScript, SQL, REST APIs, LangChain, RAG, pgvector, Prompt Engineering, Cursor IDE Agents, GitHub Copilot, AWS (Lambda, S3, EC2, API Gateway), Docker, Helm, Jenkins, Maven, SonarQube, BlackDuck, Selenium, Agile/Scrum. Open to Full stack Software Engineer / Backend Engineer roles in financial services, insurance, and healthcare. Happy to connect with recruiters and hiring managers hiring for Java, cloud, modernization, or GenAI-in-production work.

Experience

Tata Consultancy Services (Client: TSYS)

Modernization Engineer

January 2025 - Present (1 year 8 months)

Alpharetta, Georgia, United States

- Converted 50+ legacy COBOL programs into Java Spring Boot microservices with RESTful APIs, orchestrating data migration across Db2, IMS, and VSAM/ QSAM stores with 98% functional accuracy across 200+ business rules
- Built custom AI agents in Cursor IDE that ingest legacy COBOL documentation and parse business rule specifications — reducing manual code analysis from 8 hours to under 2 hours per module (75% faster)
- Optimized agentic workflows through prompt engineering, context windowing, and chunked retrieval patterns — cutting LLM token usage by 55% per cycle while improving first-pass code acceptance from 65% to 92%
- Deployed containerized microservices via Docker and Helm to AWS EKS; leveraged Lambda for event-driven processing, API Gateway for routing, S3 for artifact storage, and Step Functions for multi-stage pipeline orchestration — reducing release cycles from 2 weeks to 3 days
- Built Jenkins and GitLab CI/CD pipelines using trunk-based development with feature branch and PR gating; integrated Apache Airflow for workflow orchestration across dev, staging, and production environments
- Collaborated cross-functionally in Agile sprints translating complex legacy business rules into cloud-native microservices; mentored 2 junior engineers on agentic development practices

Technologies: Java, Spring Boot, COBOL, AWS (EKS, Lambda, S3, API Gateway, Step Functions), Docker, Helm, Jenkins, GitLab CI/CD, Airflow, Cursor IDE, GitHub Copilot, Db2, IMS, VSAM

Tata Consultancy Services

Software Engineer

November 2021 - Present (4 years 10 months)

United States

- Built full-stack web applications using JavaScript (React, Node.js, Express, MongoDB) with RESTful APIs, JWT-based authentication, and Python backend services
- Engineered end-to-end CI/CD pipelines with Jenkins, Docker, and Maven for automated deployment to AWS EC2, reducing deployment time by 60%

- Developed automated testing frameworks using Selenium and Postman, achieving 80% test coverage and eliminating 15+ hours/week of manual regression testing
- Fine-tuned an image classification model using Python and Fast.ai, achieving 92% accuracy and documenting the process as a reusable training guide for the team

Technologies: JavaScript, React, Node.js, Express, MongoDB, Python, Fast.ai, Jenkins, Docker, Maven, AWS EC2, Selenium, Postman

Citi

Application Security Engineer

July 2022 - January 2025 (2 years 7 months)

- Led vulnerability management across 15+ application teams, improving remediation efficiency by 20% and reducing mean-time-to-patch by 1 month through cross-functional collaboration
- Developed a Python script to fetch and correlate data from the NVD (National Vulnerability Database), proactively identifying prospective vulnerabilities before they surfaced in BlackDuck and Snyk — reducing exposure windows by weeks
- Integrated automated scanning into CI/CD pipelines (Jenkins, SonarQube, BlackDuck), reducing manual review effort by 25% across the application portfolio
- Built executive-level security dashboards in Tableau translating vulnerability data into actionable KPIs, informing remediation priorities for 50+ enterprise applications

Technologies: Python, BlackDuck, Snyk, SonarQube, Jenkins, Tableau, NVD API

Education

University of Texas at Arlington

Bachelor's degree, Management Information Systems, General · (2019 - 2021)

North Lake College

Associate of Science - AS, Business Administration and Management,
General · (2017 - 2018)

North Lake College

Associate of Arts and Sciences - AAS, Business Administration and
Management, General

Amity College

High School · (2013 - 2015)